Data Integration and Warehousing

Project Title:

Online Education Platform (Udemy Courses Analysis)

Project Scenario :

,Analyze Udemy online courses to identify the factors affecting course popularity ratings, subscribers, and revenue

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Analyze Udemy online courses to identify the factors affecting course popularity and revenue. The project compares free and paid courses and examines the impact of price, subscribers, reviews, and ratings on course performance.

Objectives:

- **Analyze Udemy online courses.**
- **Compare free and paid courses.**
- **Study factors affecting course popularity and revenue.**
- **Prepare data for integration and analysis.**

Data Source Description Table:

Source Name	Type	Format	Records	Notes
Udemy Courses	Dataset	CSV	3,678	Main course catalogue
Udemy Courses (Revenue & Course Analysis)	Dataset	CSV	3,676	Revenue and pricing data
Udemy Courses (hossaingh)	Dataset	CSV	209,735	Large-scale entry-level project sheet

• Unstructured Data Analysis Table:

The following table identifies which attributes in the datasets are structured and which are unstructured, along with justification for each classification.

Attribute	Data Type	Structured / Unstructured	Justification
course_id	Integer	Structured	Numeric unique identifier
title	String	Structured	Short fixed text
description	Text	Unstructured	Free text with no fixed format
headline	Text	Unstructured	Marketing text not standardized
instructional_level	String	Structured	Predefined category
price	Float	Structured	Numeric monetary value

• Transformation Plan Table:

Step	Technique	Expected Structured Output
1	Text Cleaning	Clean course description
2	Tokenization	Separated meaningful words
3	Parsing	Extracted attributes
4	Data Type Casting	Correct numeric formats
5	Normalization	Unified values
6	Validation	Accurate structured record

• Proposed Structured Output Table:

Afterapplying thetransformationplan,theresultingstructured schema will contain the following fields:

Field Name	Data Type	Source	Description
course_id	INT	Original dataset	Unique course identifier
course_title	VARCHAR	Title	Course name
course_description	TEXT	description (cleaned)	Cleaned course description
level	VARCHAR	instructional_level	Course level
price	DECIMAL	price	Course price
rating	DECIMAL	calculated field	Course rating

Lab 3: Data Cleaning & Preprocessing

1- Data Quality Assessment Table: Identified quality issues across the datasets.

Attribute	Problem Type	Issue Description	Impact	Source
price	Semantic	Some courses have price = 0	Affects revenue analysis	Udemy Courses Dataset
level	Format inconsistency	Different formats (Beginner / Beginner Level)	Inconsistent grouping	Udemy Courses Dataset
subject	Structural	Multiple categories with inconsistent naming	Difficult categorization	Udemy Courses Dataset
published_timestamp	Format inconsistency	Date format varies	Difficult time analysis	Udemy Revenue Dataset
rating	Integrity violation	Missing or inconsistent rating values	Affects evaluation	Udemy Revenue Dataset
content_duration	Structural	Different units or formats (hours/text)	Hard to compare courses	Udemy Revenue Dataset
Free/Paid	Semantic	Duplicate meaning with price column	Redundant data	Entry Level Project Sheet
Date	Format inconsistency	Different date format from timestamp	Integration issues	Entry Level Project Sheet
Subscribers	Integrity violation	Extremely high or low values	May affect analysis accuracy	Entry Level Project Sheet

2- Profiling Summary Table: Analyzed null values, duplicates, pattern issues, and key candidates.

Dataset 1: Udemy Courses Dataset

Attribute	Null %	Duplicates	Pattern Issues	Key Candidate
course_id	0%	No	No pattern issues detected	**Yes**
course_title	0%	No	Variations in text length and naming conventions	No
url	0%	No	Inconsistent URL structures and formats	No
price	0%	No	Presence of zero values indicating free courses	No
num_subscribers	0%	No	Highly skewed distribution of values	No
num_reviews	0%	No	Presence of very low or zero values	No

Dataset 2: Udemy Revenue Dataset

Attribute	Null %	Duplicates	Pattern Issues	Key Candidate
course_id	0%	No	No pattern issues detected	**Yes**
course_title	0%	No	Variations in text length and naming conventions	No
url	0%	No	Inconsistent URL structures and formats	No
price	0%	No	Presence of zero values indicating free courses	No
num_subscribers	0%	No	Highly skewed distribution of values	No
rating	0%	No	Variations in rating values	No

Dataset 3: Entry Level Project Sheet

Attribute	Null %	Duplicates	Pattern Issues	Key Candidate
course_id	0%	No	No pattern issues detected	**Yes**
course_title	0%	No	Variations in text length and naming conventions	No
url	0%	No	Inconsistent URL structures and formats	No
price	0%	No	Presence of zero values indicating free courses	No
num_subscribers	0%	No	Highly skewed distribution of values	No
Free/Paid	0%	No	Redundant representation with price	No

3- Cleaning Strategy Table:Applied cleaning techniques to improve data consistency and quality.

Issue	Cleaning Technique	Rule Applied	Justification	Expected Outcome
Zero price values	Categorization	Keep as free courses	Reflect real scenario	Accurate analysis
Level inconsistency	Standardization	Convert to unified labels	Consistency	Better grouping
Date format mismatch	Convert format	Use YYYY-MM-DD	Easy integration	Unified dates
Redundant columns	Attribute reduction	Drop Free/Paid or derive from price	Avoid redundancy	Clean schema
Rating issues	Data validation and correction	Identify invalid values and correct or remove them	Improve reliability	Accurate metrics

Lab 4: Schema Matching:

• Schema Matching:

Schema matching was performed between the Udemy Courses Dataset and the Udemy Revenue Dataset to identify corresponding attributes before data integration.

• Matching Results:

High Match: course_title, price, num_subscribers, num_reviews, num_lectures
Partial Match: level, content_duration, subject.

• Confidence Levels:

High Confidence → Same name , meaning or data type.
Medium Confidence → Same concept but different format or values.

Source A Attribute	Source B Attribute	Match Type	Confidence	Justification
course_title	course_title	Match	High	Same name and same meaning (course title)
price	price	Match	High	Same name and same data type (numeric price)
num_subscribers	num_subscribers	Match	High	Same name and meaning (number of subscribers)
num_reviews	num_reviews	Match	High	Same name and meaning (number of reviews)
num_lectures	num_lectures	Match	High	Same name and meaning (number of lectures)
level	level	Partial Match	Medium	Same concept but values may differ (e.g., Beginner vs All Levels)
content_duration	content_duration	Partial Match	Medium	Same meaning but may differ in format (numeric vs string)
subject	subject	Partial Match	Medium	Same concept but category values may differ

Lab 5: Schema Mapping :

• Schema Mapping:

Following the schema matching phase, we established structured schema mapping rules to define how data transforms and flows from the source datasets into a unified target schema, ensuring data consistency and readiness for the warehouse integration.

• Mapping Types Used:

Direct Mapping(1:1):
Used for attributes with identical definitions across source and target schemas (e.g., course_title , price).
Conditional Mapping:
Usedtotransform and standardizeinconsistent values (e.g., level).

• Schema Mapping Table:

Source Attribute(s)	Target Attribute(s)	Mapping Type	Short Description
course_title	course_title	1:1	Direct mapping as both attributes have the same name and meaning
price	price	1:1	Same numeric attribute representing course price
level	level	Conditional	Same concept but values may differ (e.g., Beginner vs All Levels)

• SQL Transformation Rules:

Rule	Rule Expression (SQL)	Explanation
1	SELECT course_title AS course_title FROM Source;	Direct mapping of course title since both attributes are identical
2	SELECT price AS price FROM Source;	Direct mapping of numeric price values
3	SELECT CASE WHEN level = 'Beginner Level' THEN 'Beginner' ELSE level END AS level FROM Source;	Standardize level values to ensure consistency

Lab 6: Data Warehouse Design & ETL

• Star Schema Architecture:

The datawarehouseisdesignedusingamulti-dimensional Star Schema to isolate quantitative metrics from descriptivecontexts.

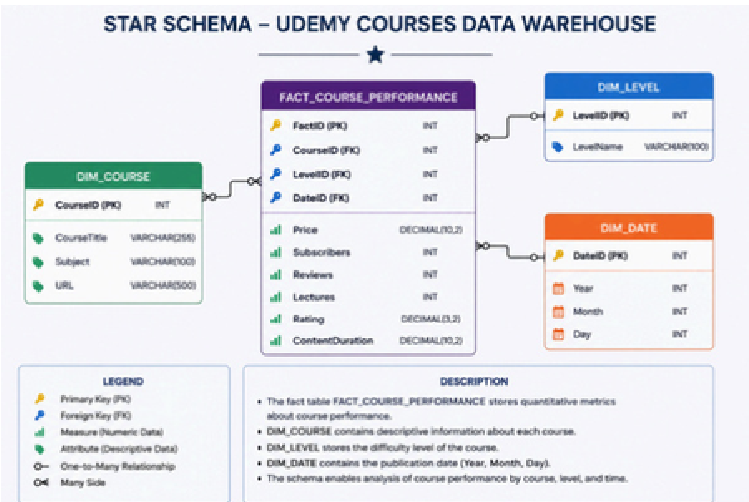
• Star Schema Design Table:

Table Type	Table Name	Keys	Main Attributes	Description
Fact	FACT_COURSE_PERFORMANCE	FactID (PK), CourseID (FK), LevelID (FK), DateID (FK)	Price, Subscribers, Reviews, Lectures, Rating, ContentDuration	Stores quantitative course performance measures
Dimension	DIM_COURSE	CourseID (PK)	CourseTitle, Subject, URL	Stores course descriptive information
Dimension	DIM_LEVEL	LevelID (PK)	LevelName	Stores course difficulty levels
Dimension	DIM_DATE	DateID (PK)	Year, Month, Day	Stores publication dates

• ETL Process Summary:

-Extraction (E): Pulling data from 3 Kaggle Udemy sources.
-Transformation (T): Dropping duplicates, standardizing level names, and parsing date segments (Year, Month, Day).
-Loading (L): Populating Dimension tables first, followed by the Fact table.

Schema Diagram:



• Data Loading Log:

Table Name	Rows Inserted	Source(s) Used	Notes / Simplifications
DIM_COURSE	3	Dataset 1, Dataset 2, Dataset 3	Loaded representative course profiles, cleaned titles, and mapped to unique IDs.
DIM_LEVEL	3	Dataset 1, Dataset 2	Normalized unique curriculum difficulty levels (Beginner, Intermediate, Advanced).
DIM_DATE	3	Dataset 1, Dataset 2	Parsed distinct chronological segments from raw publication timestamps.
FACT_COURSE_PERFORMANCE	3	Dataset 1, Dataset 3	Aggregated metrics including prices, subscribers, and lectures, and validated all operational foreign keys.

Lab 7: Data Warehouse Queries & Analytics

Inthislab,weperformedanalytical queriesonourdata warehouse to extract key businessinsightsregardingcourse performanceanduser engagement.

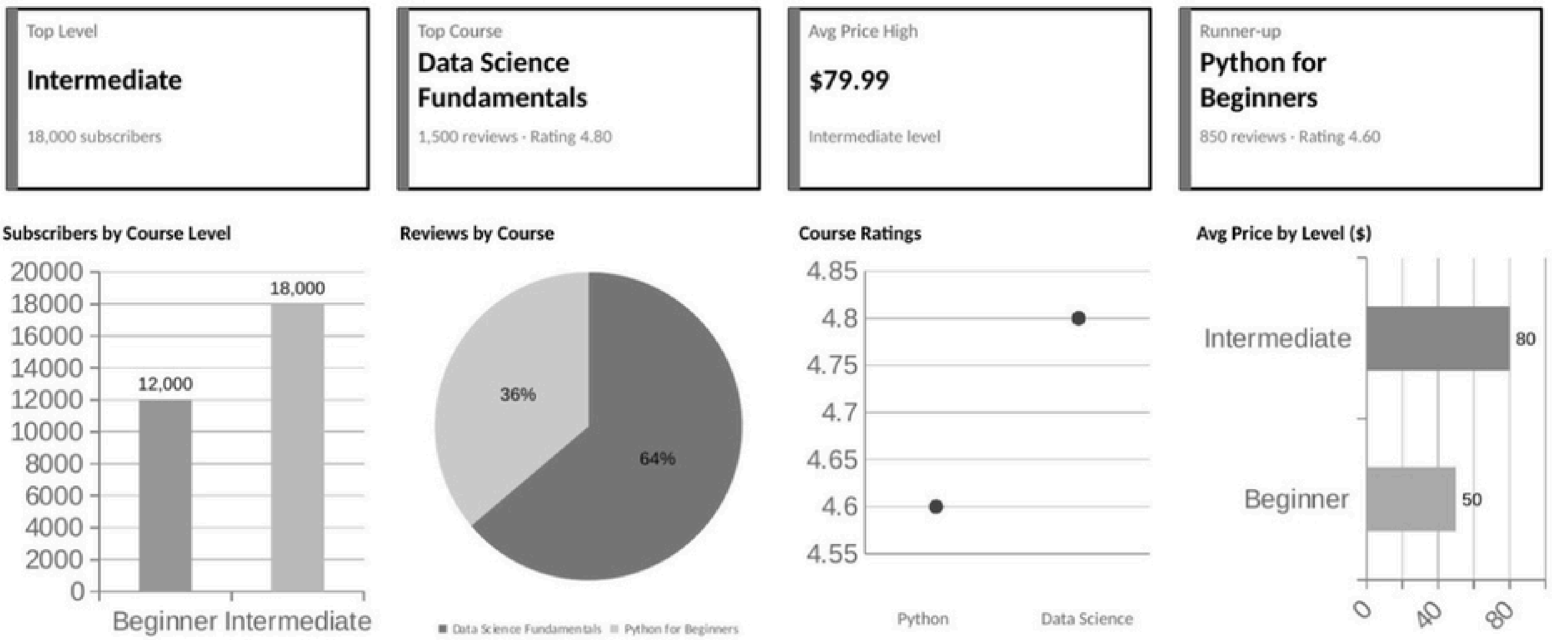
- ## Template A - Query Log:

Query #	Business Question	Tool (SQL / Report)	Result Summary
1	Which course level has the highest number of subscribers?	SQL + Google Colab	Intermediate level courses have the highest number of subscribers.
2	Which course generates the highest number of reviews?	SQL + Google Colab	Data Science Fundamentals generated the highest number of reviews.
3	Which course has the highest rating?	SQL + Google Colab	Data Science Fundamentals achieved the highest course rating.
4	Which course has the highest averge price?	SQL + Google Colab	I ntermediate level courses have the highest average price.

- ## SQL Analysis & Visualizations:

Query	Question	SQL Query	Results
Q1	Which course level has the highest number of subscribers?	SELECT I.LevelName, SUM(f.Subscribers) AS TotalSubscribers FROM FACT_COURSE_PERFORMANCE f JOIN DIM_LEVEL I ON f.LevelID = I.LevelID GROUP BY I.LevelName ORDER BY	LevelName / TotalSubscribers Intermediate / 18000 Beginner / 12000
Q2	Which course generates the highest number of reviews?	SELECT c.CourseTitle, f.Reviews FROM FACT_COURSE_PERFORMANCE f JOIN DIM_COURSE C ON f.CourseID = c.CourseID	CourseTitle / Reviews Data Science Fundamentals / 1500 Python for Beginners / 850
Q3	Which course has the highest rating?	SELECT c.CourseTitle, f.Rating FROM FACT_COURSE_PERFORMANCE f JOIN DIM_COURSE C ON f.CourseID = c.CourseID	CourseTitle / Rating Data Science Fundamentals / 4.80 Python for Beginners / 4.60
Q4	Which course level has the highest average price?	SELECT I.LevelName, AVG(f.Price) AS AveragePrice FROM FACT_COURSE_PERFORMANCE f JOIN DIM_LEVEL I ON f.LevelID = I.LevelID GROUP BY I.LevelName	LevelName / AveragePrice Intermediate / 79.990000 Beginner / 49.990000

Key Insights & Visual Dashboards:



- **Highest Subscribers: Intermediate Level (18,000).**
- **Highest Reviews: Data Science Fundamentals (1,500).**
- **Highest Rating: Data Science Fundamentals (4.80).**
- **Highest Average Price: Intermediate Level ($79.99).**

Final
Recommendations & Lessons Learned

- ## Lessons Learned:

-Integration Challenge:
Converting raw datasets from different sources into a unified warehouse was the most complex part of our project.

-Design is Critical:
Every decision made during the schema mapping phase (Lab 5) directly determined the efficiency of our final queries (Lab 7).

-ETL Workflow:
We learned that cleaning, transforming, and loading data in a structured sequence is essential to avoid data inconsistencies.

- ## Recommendations:

-Automate ETL:
Instead of manual cleaning, implementing automated scripts would save time and reduce errors in future projects.

-Expand Data Sources:
The current Star Schema is scalable; we recommend adding more datasets to gain deeper business insights.

-Interactive Reporting:
For better visualization, we suggest replacing static query results with interactive dashboards for real-time tracking.